

Computer Networks Assignment - 01

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Semester:- 4th

Q.1 $\Rightarrow$ Transmission speed = 20 bps.
Propagation speed = 180 m/ms.

$$\therefore \text{Transmission time} = \frac{1}{\text{Transmission Speed}} \Rightarrow \frac{1}{20} \Rightarrow 0.05 \text{ sec}$$

$\Rightarrow$ ~~Distance covered in one second~~ $180 \times 10^3 \text{ m}$

$$\therefore \text{1-bit delay} = \frac{\text{Transmission time} \times \text{Distance covered in one second}}{1}$$

$$\Rightarrow 0.05 \times 180 \times 10^3$$

$$\Rightarrow \text{Transmission time} \Rightarrow 0.05 \text{ sec}$$

$$\text{* Propagation Speed} \Rightarrow 180 \text{ m/ms.}$$

$$\therefore 1 \text{ bit-delay} \Rightarrow 180 \times 0.5 \Rightarrow 9 \text{ m}$$

$\Rightarrow$ Therefore the 1-bit delay in this network is equivalent to approx 9m of cable.

(d)
Ques. 2 $\rightarrow$ length of packet = 8000 bits
 Bandwidth = 400 Mbps = 400×10^6 bps
 (BW)

Speed of Signal = 250000 Km/s $\Rightarrow 2.5 \times 10^5$ Km/s
 (v)

* let $d \rightarrow$ length of ethernet cable.

$\Rightarrow T_p \rightarrow$ propagation time
 $T_t \rightarrow$ transmission time.

$\Rightarrow$ For CSMA/CD (Carrier sense Multiple Access with Collision detection):

$\Rightarrow$ ~~Maximum cable length = $\frac{\text{Data rate} \times \text{round trip time}}{\text{frame size} \times \text{utilization} \times \text{Speed of signal}}$~~

* ~~Round trip time~~

* $T_t \geq 2 \times T_p$

* Therefore; $\frac{L}{BW} \geq 2 \times \frac{d}{v}$

$\Rightarrow d \leq \frac{L \times v}{2 \times BW}$

* But utilization is 50% (0.5), therefore

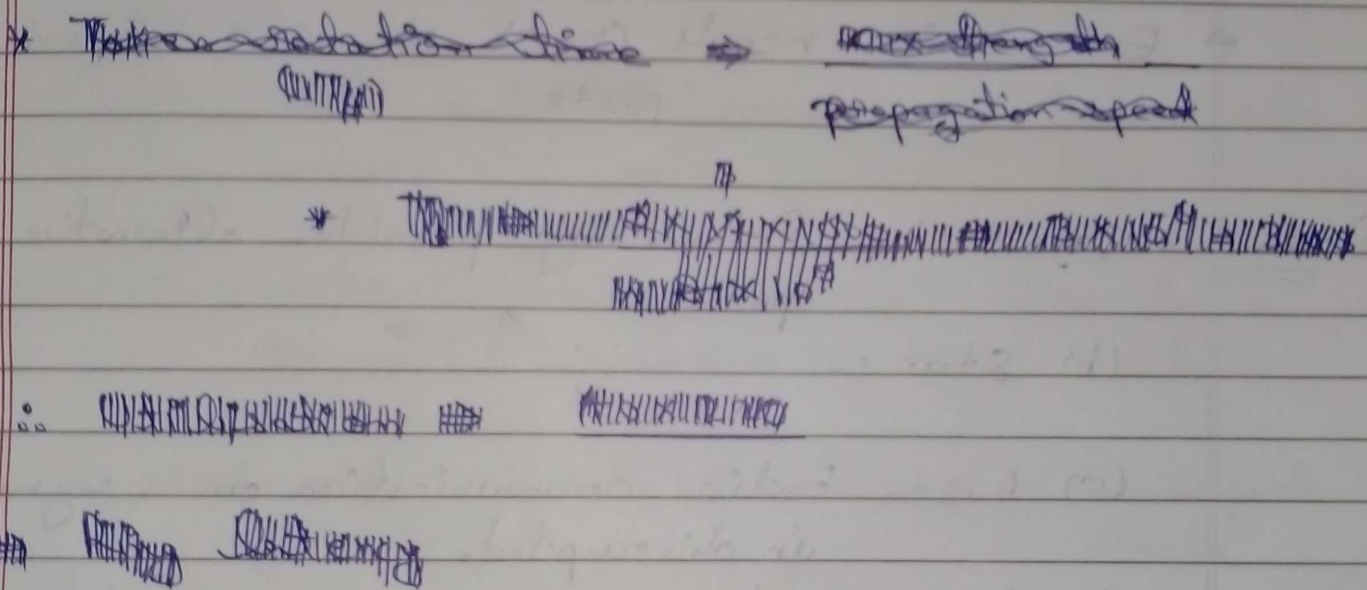
$\Rightarrow d \leq \frac{L \times v}{2 \times BW \times 0.5}$

$\Rightarrow d \leq 5 \text{ Km} \quad \therefore \boxed{d_{\max} = 5 \text{ Km}} \quad \underline{\text{Ans}}$

Q.3 $\Rightarrow$ Length of ring $\Rightarrow 12 \text{ km}$
 Stations $\Rightarrow 40$
 propagation speed $\Rightarrow 8 \text{ ns}$
 Data rate (transmission speed) $\Rightarrow 100 \text{ Mbps}$

frame size $\Rightarrow 220 \text{ bytes}$

$\Rightarrow$ Delay occurred at each station $\Rightarrow$ is equivalent to 10-bit delay



$$\Rightarrow \text{Efficiency} \Rightarrow \frac{N \times T_t}{T_p + (THT \times N)}$$

where $N = \text{No. of Stations} = 40$

$$T_t = \text{transmission time (delay)} = \frac{220 \times 8 \text{ bits}}{100 \times 10^6 \text{ bits/s}} \Rightarrow 1760 \text{ ns}$$

$$T_p = \text{propagation delay} \Rightarrow \frac{\text{Length}}{\text{Prop. Speed}} \Rightarrow \frac{12 \times 1000}{8 \times 10^9} \Rightarrow 1.5 \text{ ns}$$

$$\Rightarrow THT (\text{token holding time}) \Rightarrow T_t + T_p \Rightarrow 1761.5 \text{ ns}$$

$$\Rightarrow \text{Efficiency} = \frac{40 \times 1760 \times 10^{-6}}{(1.5 + 70460) \times 10^{-6}} \Rightarrow 0.9991$$

$\therefore$ Utilization $\Rightarrow 99.91\%$ Ans

Ques. 4 $\Rightarrow$ No. of devices = 6 (in mesh topology)

$$\therefore \text{Cables required} \Rightarrow \frac{6(6-1)}{2} \Rightarrow 15$$

$\Rightarrow$ Each device needs $(n-1)$ $\Rightarrow 6-1 \Rightarrow 5$ ports.

Ques. 5 $\Rightarrow$ (a) Mesh:- Redundancy provides alternative path.

(b) Star:-

(c) Bus:- Entire communication on a segment is disrupted.

(d) Ring:- Entire Network communication halts until the connection is restored.

Ques. 6 $\Rightarrow$ Applications ^{more} sensitive to delay:-

(i) Sending an email.

(ii) Surfing the internet.

Ques. 7 $\Rightarrow$ Local telephone calls are point to point connections because they involve only two parties communicating directly.

Ques. 8 ⇒ (a) Route determination :- Network layer.

(b) Flow Control :- Data link layer
Transport layer.

(c) Interface to transmission media :- Physical layer.

(d) Provides access for the end user :- Application layer.

Ques. 9 ⇒ (a) Format and code conversion :- Presentation
Services layer.

(b) Establishes, manages and terminates sessions :- Session layer.

(c) Ensures reliable transmission of data :- Transport layer.

(d) log-in and log-out procedures :- Session layer.

(e) Provides independence from differences in data representation :- Presentation layer.

Ques. 10 ⇒ If the physical destination address of a frame is corrupted on a bus topology LAN, the frame is usually discarded. The sender can be informed through Acknowledgement mechanisms or error detection protocols.

Ques. 11 $\Rightarrow$ If the logical destination address of a packet is corrupted at the network layer, routers may not be able to route it not corrected. Source computer can be informed through ICMP message.

Ques. 12 $\Rightarrow$ If there is no process with the destination port address running at the destination computer, the receiving computer typically sends a TCP RST packet to inform the sender that the port is not available.

Ques. 13 $\Rightarrow$ From Shannon-Hartley theorem:-

$\rightarrow$ Shannon Capacity gives the theoretical maximum data rate or capacity of a noisy channel:-
(in bps)

$$\Rightarrow \text{Capacity} = \text{Bandwidth} \times \log_2 (1 + \text{SNR})$$

$\downarrow$
[Signal-to-Noise Ratio]

$$\begin{aligned} \Rightarrow \text{Capacity} &= (1 \text{ MHz}) \times \log_2 (1 + 63) \\ &= 1 \times 10^6 \times \log_2 (64) \\ &\Rightarrow 6 \times 10^6 \Rightarrow 6 \text{ Mbps} \end{aligned}$$

$\therefore$ Appropriate bit rate = 6 Mbps.

* From Nyquist theorem (for no. of signal levels):-

$\therefore$ from this theorem:-

$$C = 2 \times \text{Bandwidth} \times \log_2 (M)$$

$\downarrow$ (Shannon Capacity) $\downarrow$ No. of discrete signal levels.

$$\Rightarrow \log_2(M) = \frac{6 \times 10^6}{2 \times 1 \times 10^6} \Rightarrow 3$$

$$\therefore M = 2^3 \Rightarrow 8 \text{ discrete levels}$$

$\therefore$ The signal level corresponds to 8 discrete signal levels.

Ques. 14 $\Rightarrow$ SNR ≈ 0
let Bandwidth = B

$$\begin{aligned} \therefore \text{Shannon Capacity} &= \text{Bandwidth} \times \log_2(1 + \text{SNR}) \\ &= B \times \log_2(1 + 0) \\ &\approx B \times \log_2(1) \\ * \text{Capacity} &\approx 0 \quad (\because \log_2 1 = 0). \end{aligned}$$

$\therefore$ For this extremely noisy channel, the channel capacity is nearly negligible due to the overwhelming noise interference.

Ques. 15 $\Rightarrow$ Given ring latency $\Rightarrow 150 \text{ ms}$
frame size $\Rightarrow 200 \text{ bytes}$
Token ring LAN speed $\Rightarrow 3 \text{ Mbps}$.

$$\begin{aligned} * \text{Transmission time for one-frame} &\Rightarrow \frac{\text{Frame Size}}{\text{LAN Speed}} \\ &\Rightarrow \frac{200 \times 8 \text{ bits}}{3 \times 10^6 \text{ bits/sec}} \\ &\Rightarrow 533.33 \mu\text{s} \end{aligned}$$

$$\begin{aligned} \Rightarrow \text{Total time taken by a frame to circulate around the ring} &\Rightarrow \text{Transmission time} + \text{Latency} \\ &\Rightarrow 533.33 \mu\text{s} + 150 \text{ ms} \\ &\Rightarrow 150.533 \text{ milliseconds} \end{aligned}$$

* Effective data rate = $\frac{200 \times 4}{150.533 \text{ ms}}$

$\Rightarrow \frac{800}{150.533 \text{ ms}} \Rightarrow 6.64 \text{ frames per sec.}$

$\Rightarrow$ Since one frame = 200 byte $\Rightarrow 200 \times 8 \text{ bits.}$

$\therefore$ Effective data rate = $200 \times 8 \times 6.64 \text{ bps}$
 $\Rightarrow 10624 \text{ bps.}$